MIL-C-45953(WC)
AMENDMENT 3
11 June 1976
SUPERSEDING
AMENDMENT 2
18 December 1975

MILITARY SPECIFICATION

CONTROL PANEL, PILOT

This amendment forms a part of Military Specification MIL-C-45953(WC), dated 22 January 1974.

Page 1

Paragraph 2.1 - Under "SPECIFICATIONS", add the following specification:

"MIL-A-45967 - Armament Subsystem, Helicopter: M28A1."

Under "STANDARDS", add the following standard:

"MIL-STD-781 - Reliability tests: Exponential Distribution."

Page 4

Add the following paragraphs:

"3.6 Reliability. When specified (see 6.2), the pilot control panel shall have a minimum acceptable Mean-Mission-Between-Failures (MMBF) of not less than 1085 missions with a desired and specified MMBF of 1627 missions. A mission is defined as those functions of the pilot control panel necessary to simulate the expenditure of 4000 rounds of 7.62mm ammunition and 283 rounds of 40mm ammunition during a two hour flight. A failure will be defined as a non-occurrence of the appropriate event when power is applied to the control panel and the required switches are actuated. Testing shall be as specified in 4.4.

3.6.1 Reliability criteria. The above reliability requirements are based on a minimum acceptable Mean-Time-Between-Failure (MTBF) of 2170 hours.

FSC 1090

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3.7 Environmental conditions. When specified (see 6.2), the pilot control panel shall be capable of satisfying the environmental conditions in accordance with MIL-A-45967. When subjected to the environmental conditions, the pilot control panel shall not exhibit any detrimental damage or deterioration which would prevent acceptable preformance. Testing shall be as specified in 4.4.

3.8 Electromagnetic interference (EMI). When specified (see 6.2), the pilot control panel shall be capable of satisfying the EMI requirements in accordance with MIL-A-45967. Testing shall be as specified in 4.4."

Page 5

Paragraph 4.4, line 6 - Delete "and such other inspection as is necessary to determine compliance with the requirements of the contract", and substitute "and, when specified (see 6.2), the reliability test (see 3.6 and 4.6.2), environmental conditions test (see 3.7 and 4.6.3), electromagnetic interference (EMI) test (see 3.8 and 4.6.4), and such other inspection as is necessary to determine compliance with the requirements of the contract."

Page 7

Paragraph 4.6.1, Delete "IEL 11687087" and substitute "IEL 11687987".

Add the following paragraphs:

"4.6.2 Reliability test. When specified (see 6.2), the reliability test shall be performed by repeating the test cycle defined in 4.6.2.1. For each test cycle, the failure versus test cycle data shall be cumulatively plotted. Testing shall be terminated whenever a decision point, as defined in Figure 1, has been reached.

4.6.2.1 Reliability test cycle. One test cycle (representing the functions necessary to perform one mission) shall be completed by performing the test cycle specified in 4.6.1. Procedures to verify requirements for the Master Arm Switch (3.3.3.1) and Weapon Select Switch (3.3.3.2) shall be repeated five times during each test cycle.

4.6.2.2 The contractor shall provide replacement parts as required to complete the test.

4.6.2.3 At the option of the contractor, the test may be interrupted at the end of a cycle and the test restarted at the point at which it was interrupted.

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4.6.2.4 Complete, accurate records shall be kept for each reliability test, showing each failure and part replacement, including an indication of when each failure occurred. In addition, complete, accurate records shall be kept for all time losses attributable to the panel, including the time for each stoppage.

4.6.3 Environmental conditions test. When specified (see 6.2), tests shall be performed for each environmental condition in accordance with the applicable method and procedure specified in MIL-A-45967. The control panel shall be operated during these tests in accordance with the functional test method of 4.6.1. Open areas of the control panel shall be covered to simulate aircraft installation for the humidity, salt, sand and dust tests.

4.6.4 Electromagnetic interference (EMI) test. When specified (see 6.2), the EMI test shall be performed in accordance with the EMI requirements specified in MIL-A-45967. The control panel shall be operated during the EMI test in accordance with the functional test method of 4.6.1."

Page 8

Paragraph 6.2.1 - Add the following:

"i. Reliability, environmental conditions and electromagnetic interference requirements and tests, when required (see 3.6, 3.7, 3.8, 4.4, 4.6.2, 4.6.3 and 4.6.4)."

Page 10

At top of page, delete the following:

"M28A1 SUBSYSTEM:
PILOT CONTROL PANEL"

At top of page, add as last line:

"MIL-STD-781 Test Plan VII"

Custodian:
Army - WC

Preparing Activity:
Army - WC

Project No. 1090-A042

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